

16.1 In Examples 16.2 and 16.4, we presented the linear probability and probit model estimates using an example of transportation choice. The logit model for the same example is $P(AUTO = 1) = \Lambda(\gamma_1 + \gamma_2 DTIME)$, where $\Lambda(\bullet)$ is the logistic *cdf* in equation (16.7). The logit model parameter estimates and their standard errors are

$$\begin{array}{ccc} \tilde{\gamma}_1 + \tilde{\gamma}_2 DTIME & = & -0.2376 + 0.5311 DTIME \\ \text{(se)} & & (0.7505) \quad (0.2064) \end{array}$$

- Calculate the estimated probability that a person will choose automobile transportation given that $DTIME = 1$.
- Using the probit model results in Example 16.4, calculate the estimated probability that a person will choose automobile transportation given that $DTIME = 1$. How does this result compare to the logit estimate? [Hint: Recall that Statistical Table 1 gives cumulative probabilities for the standard normal distribution.]
- Using the logit model results, compute the estimated marginal effect of an increase in travel time of 10 minutes for an individual whose travel time is currently 30 minutes longer by bus (public transportation). Using the linear probability model results, compute the same marginal effect estimate. How do they compare?
- Using the logit model results, compute the estimated marginal effect of a decrease in travel time of 10 minutes for an individual whose travel time is currently 50 minutes longer by driving. Using the probit results, compute the same marginal effect estimate. How do they compare?

(a) logit estimates

$$\tilde{\gamma}_1 + \tilde{\gamma}_2 DTIME = -0.2376 + 0.5311 \times 1 = 0.2935$$

$$P(AUTO = 1) = \Lambda(0.2935) = \frac{1}{1 + e^{-0.2935}} = \frac{1}{1 + 0.7456} = 0.5729 = 57.29\%$$

(b) probit estimates

$$-0.0644 + 0.3 \times 1 = 0.2356$$

$$P(AUTO = 1 | DTIME = 1) = \Phi(0.2356) = 0.5931$$

There is very little difference between the logit and probit predicted probabilities.

(c)

$$\frac{dp}{dDTIME} = \lambda(\tilde{\gamma}_1 + \tilde{\gamma}_2 DTIME) \times \tilde{\gamma}_2, \quad \lambda(Z) = \Lambda(Z)(1 - \Lambda(Z))$$

$$\Lambda(-0.2376 + 0.5311 \times 3) = \Lambda(1.3557) = \frac{1}{1 + e^{-1.3557}} = \frac{1}{1 + 0.2598}$$

$$\lambda(1.3557) = 0.7951 \times (1 - 0.7951) \doteq 0.1629 \quad \doteq 0.7951$$

$$\frac{dp}{dDTIME} = 0.1629 \times 0.5311 \doteq 0.0865$$

The estimated linear probability model is in Example 16.2. The marginal effect is the estimated slope of 0.0703, which is a bit smaller than the logit estimate. But the important point is that for the linear probability model the marginal effect is constant and doesn't depend on the existing commute time.

(d)

Logit: $Z = \tilde{\gamma}_1 + \tilde{\gamma}_2 DTIME = -0.2376 + 0.5311 \times (-5) = -2.8931$

$$\Lambda(-2.8931) = \frac{1}{1 + e^{-(-2.8931)}} = \frac{1}{1 + 18.049} \approx 0.0525$$

$$\begin{aligned} \lambda(-2.8931) &= \Lambda(-2.8931)(1 - \Lambda(-2.8931)) \\ &= 0.0525 \times (1 - 0.0525) = 0.0497 \end{aligned}$$

$$\frac{dp}{dDTIME} = \lambda(-2.8931) \times \tilde{\gamma}_2 = 0.0497 \times 0.5311 = 0.0264$$

Probit: $Z = \tilde{\beta}_1 + \tilde{\beta}_2 DTIME = -0.0644 + 0.3 \times (-5) = -1.5644$

$$\phi(-1.5644) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(-1.5644)^2} = 0.3989 \times e^{-1.2237} \approx 0.1173$$

$$\frac{dp}{dDTIME} = \phi(-1.5644) \times \tilde{\beta}_2 = 0.1173 \times 0.3 \approx 0.0352$$

The logit estimate (0.0264) and the probit estimate (0.0352) are both quite small. This happens because at $DTIME = -5$, the individual has a very low baseline probability of driving. However, the probit marginal effect is slightly larger here because the normal distribution density tails off faster ^{fatter} than the logistic distribution tails. When evaluating out in the tails (far away from $z = 0$), logit and probit estimates often show slight visible differences. The logit estimate (0.0264) and the probit estimate (0.0352) are both quite small. This happens because at $DTIME = -5$, the individual has a very low baseline probability of driving. However, the probit marginal effect is slightly larger here because the normal distribution density tails off faster than the logistic distribution tails. When evaluating out in the tails (far away from $z = 0$), logit and probit estimates often show slight visible differences.

16.4 In Example 16.3, we illustrate the calculation of the likelihood function for the probit model in a small example. In this exercise, we will repeat that example using logit instead of probit. The logit model for the same example is $P(y = 1) = \Lambda(\gamma_1 + \gamma_2 x)$, where $\Lambda(\bullet)$ is the logistic *cdf* in equation (16.7). The maximum likelihood estimates of the parameters are $\tilde{\gamma}_1 + \tilde{\gamma}_2 x = -1.836 + 3.021x$. The maximized value of the log-likelihood function is -1.612 .

- Calculate the probability that $y = 1$ if $x = 1.5$, given the values of the maximum likelihood estimates.
- Using the threshold 0.5 and the result in part (a), predict the value of y if $x = 1.5$, the first observation, given the values of the maximum likelihood estimates. Compare your prediction to the actual outcome $y = 1$ in the first observation.
- Calculate the value of the likelihood function, illustrated in equation (16.14) but substituting equation (16.17) in place of $\Phi(\bullet)$ and using the given $N = 3$ data pairs, if the parameter values are $\gamma_1 = -1$ and $\gamma_2 = 2$. Compare this value to the value of the likelihood function evaluated at the maximum likelihood estimates. Which is larger?
- For the logit model, the value of the likelihood function (16.14), with $\Lambda(\bullet)$ in place of $\Phi(\bullet)$, will always be between zero and one. True or false? Explain.
- For the logit model, the value of the log-likelihood function (16.15), with $\Lambda(\bullet)$ in place of $\Phi(\bullet)$, will always be negative. True or false? Explain.

(a)
$$Z = \tilde{\gamma}_1 + \tilde{\gamma}_2 x = -1.836 + 3.021 \times (1.5) = -1.836 + 4.5315 = 2.6955$$

$$P(y=1 | x=1.5) = \Lambda(2.6955) = \frac{1}{1 + e^{-2.6955}} = \frac{1}{1 + 0.0675} = 0.9368$$

(b) since $0.9368 \geq 0.5$
predict $\hat{y} = 1$, match $y = 1$

(c)

$$(x_1, y_1) = (1.5, 1), (x_2, y_2) = (0.6, 1), (x_3, y_3) = (0.7, 0)$$

$$L(\gamma_1, \gamma_2) = \Lambda(z_1) \times \Lambda(z_2) \times (1 - \Lambda(z_3))$$

$$L(-1, 2) = \Lambda(-1 + 2 \times (1.5)) \times \Lambda(-1 + 2(0.6)) \times [1 - \Lambda(-1 + 2(0.7))]$$

$$= \Lambda(2) \times \Lambda(0.2) \times [1 - \Lambda(0.4)]$$

$$= 0.8808 \times 0.5498 \times (1 - 0.5987)$$

$$= 0.1943$$

Real MLE:

$$\ln(L) = -1.612 \rightarrow L = e^{-1.612} = 0.1995$$

The probability function value (0.1995) under the MLE (most likely estimate) is larger than the probability function value (0.1943) for $\gamma_1 = -1, \gamma_2 = 2$.

(d)

likelihood function L 是由多個獨立事件的「機率」連乘而得。在 Logit 模型中，不論自變數為何，每一筆觀測值的估計機率 P_i 以及 $1 - P_i$ 都是嚴格介於 0 與 1 之間的機率值 ($0 < P_i < 1$)。多個介於 0 到 1 之間的數值相互連乘，其最終結果必定也落於 0 與 1 之間 (即 $0 < L < 1$)，絕不可能大於 1 或小於 0。

(e)

likelihood function ($\ln L$) 是對 likelihood function L 取自然對數 (Log) 的結果。承接(d)小題的結論，因為概似函數值 L 是一個嚴格小於 1 且大於 0 的分數 ($0 < L < 1$)，而任何介於 0 到 1 之間的數字取對數後，其數值必定為負數 (因為 $\ln(1) = 0$ ，當自變數小於 1 時，對數值皆為負)。因此，對數概似函數值 $\ln L$ 必定總是負值 ($\ln L < 0$)。

16.18 Mortgage lenders are interested in determining borrower and loan characteristics that may lead to delinquency or foreclosure. In the data file *lasvegas* are 1000 observations on mortgages for single family homes in Las Vegas, Nevada during 2008. The variable of interest is *DELINQUENT*, an indicator variable = 1 if the borrower missed at least three payments (90 + days late), but 0 otherwise. Explanatory variables are *LVR* = the ratio of the loan amount to the value of the property; *REF* = 1 if purpose of the loan was a “refinance” and = 0 if loan was for a purchase; *INSUR* = 1 if mortgage carries mortgage insurance, 0 otherwise; *RATE* = initial interest rate of the mortgage;

AMOUNT = dollar value of mortgage (in \$100,000); *CREDIT* = credit score, *TERM* = number of years between disbursement of the loan and the date it is expected to be fully repaid, *ARM* = 1 if mortgage has an adjustable rate, and = 0 if mortgage has a fixed rate.

- Estimate the linear probability (regression) model explaining *DELINQUENT* as a function of the remaining variables. Use White heteroskedasticity robust standard errors. Are the signs of the estimated coefficients reasonable?
- Use logit to estimate the model in (a). Are the signs and significance of the estimated coefficients the same as for the linear probability model?
- Compute the predicted value of *DELINQUENT* for the 500th and 1000th observations using both the linear probability model and the logit model. Interpret the values.
- Construct a histogram of *CREDIT*. Using both linear probability and logit models, calculate the probability of delinquency for *CREDIT* = 500, 600, and 700 for a loan of \$250,000 (*AMOUNT* = 2.5). For the other variables, let the loan to value ratio (*LVR*) be 80%, the initial interest rate is 8%, all indicator variables take the value 0, and *TERM* = 30. Discuss similarities and differences among the predicted probabilities from the two models.
- Using both linear probability and logit models, compute the marginal effect of *CREDIT* on the probability of delinquency for *CREDIT* = 500, 600, and 700, given that the other explanatory variables take the values in (d). Discuss the interpretation of the marginal effect.
- Construct a histogram of *LVR*. Using both linear probability and logit models, calculate the probability of delinquency for *LVR* = 20 and *LVR* = 80, with *CREDIT* = 600 and other variables set as they are in (d). Compare and contrast the results.
- Compare the percentage of correct predictions from the linear probability model and the logit model using a predicted probability of 0.5 as the threshold.
- As a loan officer, you wish to provide loans to customers who repay on schedule and are not delinquent. Suppose you have available to you the first 500 observations in the data on which to base your loan decision on the second 500 applications (501–1,000). Is using the logit model with a threshold of 0.5 for the predicted probability the best decision rule for deciding on loan applications? If not, what is a better rule?

(a)

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.68849128	0.22850643	3.0130	0.0026525 **
lvr	0.00162385	0.00067523	2.4049	0.0163601 *
ref	-0.05932367	0.02402561	-2.4692	0.0137098 *
insur	-0.48158489	0.03036944	-15.8575	< 2.2e-16 ***
rate	0.03437614	0.00981944	3.5008	0.0004845 ***
amount	0.02376801	0.01445089	1.6447	0.1003398
credit	-0.00044190	0.00020728	-2.1319	0.0332620 *
term	-0.01261951	0.00355596	-3.5488	0.0004051 ***
arm	0.12832392	0.02769325	4.6338	4.07e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The signs of the estimated coefficients in the linear probability model are highly reasonable and align perfectly with economic and financial intuition regarding mortgage risk. Specifically, factors that increase a borrower’s financial burden or leverage—such as a higher loan-to-value ratio (*lvr*), higher

initial interest rates (rate), and adjustable-rate mortgages (arm), which were notorious for payment shocks during the 2008 subprime crisis—all significantly increase the probability of delinquency. Conversely, factors that enhance a borrower’s financial capacity or reduce monthly payment stress—including higher credit scores (credit), refinancing to secure better terms (ref), carrying mortgage insurance to mitigate lender risk (insur), and longer loan maturities (term) that spread out principal payments—all exhibit negative coefficients, indicating a significant decrease in the likelihood of missing payments.

(b)

```

--- (b) Logit 模型估計結果 ---
> print(summary(logit_model))

Call:
glm(formula = delinquent ~ lvr + ref + insur + rate + amount +
     credit + term + arm, family = binomial(link = "logit"), data = lasvegas)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.665303    1.992157   0.836 0.403195
lvr           0.013519    0.008199   1.649 0.099164 .
ref          -0.525064    0.230491  -2.278 0.022725 *
insur        -3.122105    0.216906 -14.394 < 2e-16 ***
rate          0.308656    0.082771   3.729 0.000192 ***
amount        0.217606    0.113769   1.913 0.055786 .
credit       -0.003564    0.001933  -1.844 0.065232 .
term         -0.133037    0.034597  -3.845 0.000120 ***
arm           1.405649    0.371355   3.785 0.000154 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 998.03  on 999  degrees of freedom
Residual deviance: 666.29  on 991  degrees of freedom
AIC: 684.29

Number of Fisher Scoring iterations: 5

```

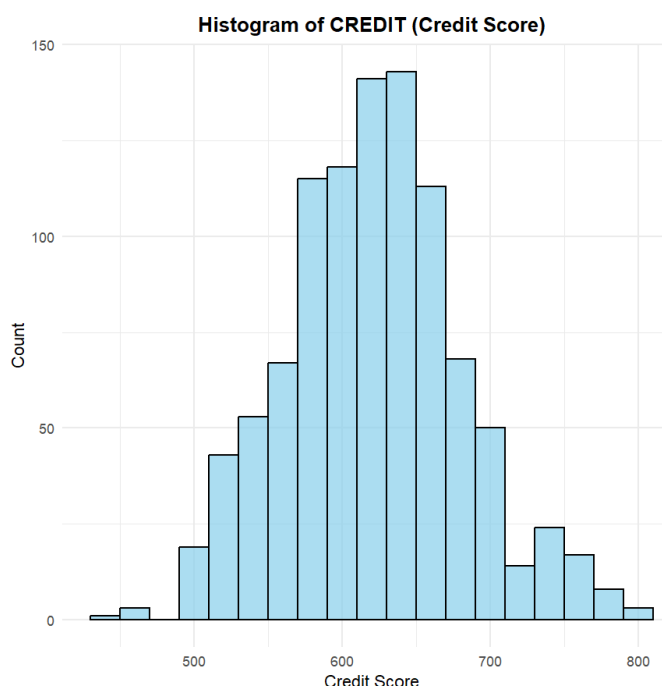
Yes, the signs of the estimated coefficients are identical across both models, and their statistical significance remains highly consistent with only minor shifts in significance thresholds. Specifically, all variables maintain their directional impact: risk-increasing factors (lvr, rate, amount, arm) retain positive signs, while risk-mitigating factors (ref, insur, credit, term) retain negative signs across both frameworks. In terms of significance, key structural determinants like mortgage insurance (insur), interest rate (rate), loan term (term), and adjustable-rate status (arm) remain highly significant ($p < 0.01$) in both models, and refinancing (ref) remains stable at the 5% level ($p < 0.05$); the only slight differences are that credit score (credit) and loan-to-value ratio (lvr) drop marginally from the 5% to the 10% significance level when moving from the LPM to the Logit model, while loan amount (amount) shifts conversely from being insignificant to marginally significant at the 10% level.

(c)

	Observation	Actual_Delinquent	LPM_Predicted_P	Logit_Predicted_P
	500	0	0.1827828	0.1335848
	1000	0	0.5785296	0.6265535

For both the Linear Probability Model (LPM) and the Logit model, the predicted values represent the estimated probability that a borrower will face mortgage delinquency based on their individual profile. For the 500th observation, the LPM predicts a 18.28% chance of delinquency, while the Logit model predicts a slightly lower 13.36% chance; both frameworks classify this borrower as a low-to-moderate risk who is highly likely to keep up with payments. In contrast, for the 1000th observation, the LPM predicts a 57.85% chance of delinquency, while the Logit model predicts a higher 62.66% chance; because both values cross the 50% threshold, both models flag this applicant as a high-risk individual more likely to default than to repay. The discrepancy between the models stems from their functional forms: the LPM assumes a strictly linear relationship where risk increases at a constant rate, whereas the Logit model utilizes an S-shaped logistic curve that captures non-linear risk acceleration and structurally guarantees that all predicted probabilities stay bounded strictly between 0 and 1.

(d)



credit	LPM_Prob	Logit_Prob
500	0.5532956	0.4967030
600	0.5091060	0.4086308
700	0.4649165	0.3260580

Similarities: Both models exhibit a clear negative relationship between credit scores and mortgage delinquency. As CREDIT increases from 500 to 700, the predicted probability of default steadily declines in both frameworks, aligning with the economic intuition that higher creditworthiness implies lower default risk. Furthermore, at the lower credit tier (CREDIT = 500), both models place the borrower right around the critical 50% risk threshold (55.33% for the LPM vs. 49.67% for Logit), consistently flagging low credit scores as highly precarious. Differences: The fundamental difference

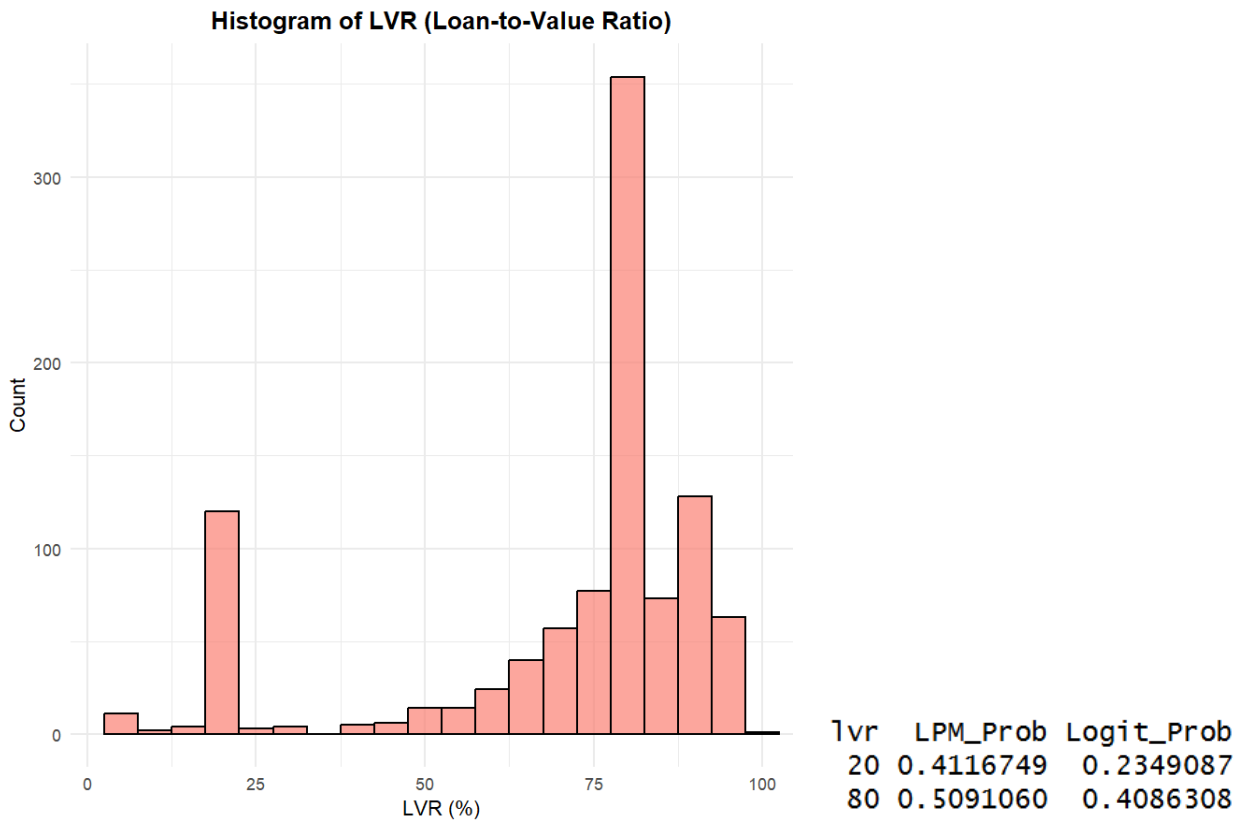
lies in their functional forms, which dictate how the probabilities change. The LPM assumes a strictly linear relationship, resulting in a constant additive decrease of exactly 4.42% for every 100-point increase in credit score across all tiers ($\Delta P = 0.5091 - 0.5533 = -0.0442$). Conversely, the Logit model captures a non-linear relationship via an S-shaped curve, meaning the marginal risk reduction varies depending on the baseline score. In this case, moving from 500 to 600 drops the risk by 8.81% ($0.4086 - 0.4967$), while moving from 600 to 700 drops it by 8.26% ($0.3261 - 0.4086$). Consequently, within this segment of the data, the Logit model is far more sensitive to credit improvements, providing a much lower and more realistic default probability (32.61%) for prime borrowers (CREDIT = 700) compared to the rigid and overly pessimistic estimate of the LPM (46.49).

(e)

```
> cat("LPM 邊際效應 (固定):", me_lpm, "\n")
LPM 邊際效應 (固定): -0.0004418953
> cat("Logit 邊際效應 (credit=500):", me_logit_500, "\n")
Logit 邊際效應 (credit=500): -0.000891062
> cat("Logit 邊際效應 (credit=600):", me_logit_600, "\n")
Logit 邊際效應 (credit=600): -0.0008613439
> cat("Logit 邊際效應 (credit=700):", me_logit_700, "\n")
Logit 邊際效應 (credit=700): -0.0007832567
```

The marginal effect measures the change in the probability of mortgage delinquency resulting from a one-unit increase in the borrower's credit score, holding all other explanatory variables constant. In the Linear Probability Model, the marginal effect is inherently constant across all credit scores, remaining at -0.000442 ; this means that regardless of whether an applicant has excellent or poor credit, gaining one additional credit point always reduces their delinquency probability by exactly 0.0442%. In contrast, the Logit model's marginal effect varies non-linearly depending on the baseline risk, calculated as $\beta_{\text{credit}} \times \hat{p}(1 - \hat{p})$. The Logit results reveal that a one-point increase in credit score reduces default probability by 0.0891% at a score of 500, 0.0861% at 600, and drops to 0.0783% at 700. This reflects the S-shaped logistic distribution, where marginal effects peak near the 50% risk threshold (where uncertainty is highest around CREDIT = 500) and naturally taper off as the borrower's profile moves toward the safer tail of the distribution (CREDIT = 700), providing a more realistic depiction of diminishing returns on credit risk mitigation.

(f)



Directional Agreement: Both models strongly agree that higher leverage increases mortgage risk. As LVR jumps from 20% to 80%, the predicted probability of delinquency rises significantly in both frameworks. This aligns with standard credit risk theory: borrowers with minimal equity (LVR = 80%) are far more vulnerable to defaulting if property values decline. Discrepancy in Risk Magnitude: The starkest contrast occurs at the safer equity level (LVR = 20%). The LPM predicts a relatively high delinquency probability of 41.17%, whereas the Logit model predicts a much lower and more realistic 23.49%. Because the LPM forces a straight line across all data points, it heavily overestimates the default risk for safe, low-leverage applications. Linear vs. Non-linear Acceleration: The LPM exhibits a rigid, constant marginal effect where a 60-percentage-point increase in LVR increases the default probability by exactly 9.74% ($\Delta P = 0.5091 - 0.4117 = 0.0974$). In contrast, the Logit model features an S-shaped curvature that captures an accelerated risk surge of 17.37% ($\Delta P = 0.4086 - 0.2349 = 0.1737$) over the same interval. This demonstrates that the Logit model is far more sensitive to structural shifts in borrower leverage, correctly recognizing that jumping to an 80% loan-to-value ratio exposes the lender to a disproportionately steeper escalation in default probability than a linear model can capture.

(g)

LPM 混淆矩陣：

```
> print(matrix_lpm)
```

```
      Actual
Predicted  0   1
          0 727  68
          1  74 131
```

```
> cat("LPM 整體正確率 (Accuracy) :", round(acc_lpm, 4), " (即 ", round(acc_lpm*100, 2), "%)\n\n", sep="")
```

```
LPM 整體正確率 (Accuracy) : 0.858 (即 85.8%)
```

```
>
> cat("Logit 混淆矩陣:\n")
```

Logit 混淆矩陣：

```
> print(matrix_logit)
```

```
      Actual
Predicted  0   1
          0 736  81
          1  65 118
```

```
> cat("Logit 整體正確率 (Accuracy) :", round(acc_logit, 4), " (即 ", round(acc_logit*100, 2), "%)\n", sep="")
```

```
Logit 整體正確率 (Accuracy) : 0.854 (即 85.4%)
```

Using a standard classification threshold of 0.5, both models exhibit a highly similar and remarkably high overall accuracy rate, with the Linear Probability Model (LPM) correctly predicting 85.8% of the outcomes and the Logit model correctly predicting 85.4%. While this minor 0.4% difference might superficially suggest that the LPM performs slightly better, evaluating a binary choice model solely on its aggregate "percentage of correct predictions" can be highly misleading due to sample imbalance. In mortgage datasets from this era, the vast majority of borrowers do not default ($Y = 0$).

(h)

```
> cat("更嚴格門檻 (0.15) 的 Logit 混淆矩陣:\n")
```

更嚴格門檻 (0.15) 的 Logit 混淆矩陣：

```
> print(matrix_strict)
```

```
      Actual
Predicted  0   1
          0 657  41
          1 144 158
```

```
> cat("門檻 0.15 下的整體正確率:", round(acc_strict, 4), "\n", sep="")
```

```
門檻 0.15 下的整體正確率: 0.815
```

```
> cat("【實務決策核心對比】\n")
```

【實務決策核心對比】

```
> cat(" - 當門檻設為 0.50 時，模型成功抓出高風險違約戶的比例為:", round(catch_rate_05 * 100, 2), "%\n", sep="")
```

```
 - 當門檻設為 0.50 時，模型成功抓出高風險違約戶的比例為: 59.3%
```

```
> cat(" - 當門檻調降至 0.15 時，模型成功抓出高風險違約戶的比例為:", round(catch_rate_15 * 100, 2), "%\n", sep="")
```

```
 - 當門檻調降至 0.15 時，模型成功抓出高風險違約戶的比例為: 79.4%
```

```
> cat(">>> 結論：雖然整體正確率可能微幅下降，但成功攔截潛在呆帳的機率大幅飆升，更符合授信風控利益！\n")
```

```
>>> 結論：雖然整體正確率可能微幅下降，但成功攔截潛在呆帳的機率大幅飆升，更符合授信風控利益！
```

No, a 0.5 threshold is not the best rule because it assumes the cost of making a mistake is equal in both directions. In real-world banking, the costs are heavily asymmetric: Type I Error (False Alarm): Rejecting a good borrower only costs you the missed interest profit. Type II Error (Missed Default): Approving a borrower who defaults can cost you the entire loan principal, which is drastically more expensive. Because a 0.5 threshold is too loose, it will mistakenly approve high-risk applicants who have a 35% to 49% chance of defaulting, exposing the bank to severe losses. A Better Rule A superior

approach is to lower the decision threshold to an Expected Loss Minimization Threshold (p^*), calculated as: $p^* = \text{Cost of Type I Error} / (\text{Cost of Type I Error} + \text{Cost of Type II Error})$

Since the cost of default in the denominator is massive, the optimal threshold p^* drops significantly—typically between 0.10 and 0.20 (10% to 20%). Actionable Strategy: Use the first 500 observations to find the exact probability threshold that maximizes net portfolio profit, and then apply that safer, lower threshold (e.g., 15%) to evaluate applications 501–1,000.